

Exercise 2: Data Preparation and Quality Auditing

Three worksheets, three captions. No memo.

Item	Details
Points	100
Data	Worksheet 1: clean/mvu_student_dim.csv and clean/mvu_enrollment_fact.csv. Worksheets 2–3: raw/students_cohort_2020, 2021 and 2022.csv, raw/program_lookup.csv, raw/financial_aid.xlsx
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 2, Ch 13, Ch 14 and Ch 15
Walkthrough	Module 2 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with Worksheets 1–3 and a visible caption on each. No memo.

Before you start: document the grain

For each file, note what one row represents and its row count. Your captions will use these numbers.

File	One row is...	Rows
students_cohort_2020/21/22.csv	one student (with duplicates)	493 / 495 / 425
program_lookup.csv	one academic program	19
financial_aid.xlsx (4 sheets)	one student, one aid year, one aid type	3,897
mvu_student_dim.csv	one student	1,400
mvu_enrollment_fact.csv	one student, one course, one term	10,147

Worksheet 1: relationships versus joins

1. Connect mvu_student_dim.csv and relate mvu_enrollment_fact.csv on student_id.
2. New worksheet. From the student table, put on Text: student_id as Count (not Count Distinct), is_graduated as Average, total_aid_awarded as Sum. Name it Worksheet 1 — Relationship.
3. Duplicate the data source. In the copy, open the physical layer and rebuild the link as an inner join on student_id.
4. Duplicate the worksheet and use Data > Replace Data Source to point it at the joined copy. Name it Worksheet 1 — Join.

Checkpoint: Graduation rate on the joined version (percentage, one decimal)?

Worksheet 2: union, clean, deduplicate, join

1. New data source. Union the three raw cohort files. Merge entry_term with start_term, and enrollment_intensity with attendance_status. Leave first_generation null for 2022.

Checkpoint: Row count immediately after the union?

2. Fix at least three quality problems, keeping each original field and building a cleaned one beside it: inconsistent categories in sex and residency; whitespace and case in program_code; undergrad_gpa stored as text, with values above 4.0. Record a before-and-after count for each.
3. Standardise student_name with UPPER(TRIM()), then remove duplicates (or aggregate to one row per student_id). Clean first, then dedupe.

Checkpoint: How many rows does deduplication remove?

4. Join program_lookup.csv on the raw program_code and count the unmatched students. Then join on your cleaned code, test inner and left, and keep the left join.
5. Create IFNULL([program_name], "Unknown Program"). Put it on Rows and student_id as Count (Distinct) on Columns, sorted descending. Name it Worksheet 2.

Checkpoint: How many students have no matching program once the code is cleaned?

Worksheet 3: fan-out and the quality audit

1. Union the four sheets of financial_aid.xlsx. Left join the result to your cleaned student table on student_id.
2. On a scratch sheet, put student_id on Text as Count and as Count (Distinct). Note the gap. Fix it by aggregating aid to one row per student before joining, or by using a relationship. Do not delete records.

Checkpoint: Row count after the left join, before any fix?

3. Build Worksheet 3 on the student table (without the fan-out): record count, distinct count, duplicate keys, null count for your cleaned GPA, and GPA min and max.

Checkpoint: Null GPAs after the type change and the above-4.0 fix, on the deduplicated table?

Your three captions

Each worksheet carries a caption: a few sentences answering its question, written while the view is in front of you. Turn it on with Worksheet > Show Caption, then double-click below the view to edit. Where a worksheet has two versions, put the caption on the second one.

Sheet	What your caption must answer
Worksheet 1	What happened to the count, the graduation rate and total aid when you switched from a relationship to a join? Why did the rate go up?
Worksheet 2	The union row count and why it exceeds the number of students. Three quality problems, each with a before-and-after count. Why the raw-code join left so many unmatched, and why you kept a left join.
Worksheet 3	Is the data ready for analysis? Name the fan-out and how you fixed it without deleting records, and each remaining limitation with how it affects interpretation. End with a line confirming the data is cleared for educational use.

Worksheet 3's caption carries 25% on its own. "The data is clean" does not earn full marks. The verdict is the point of the exercise; the other two assess whether you can operate the software.

Before you submit: open each sheet and confirm the caption is showing. A caption toggled off still holds your text but won't be visible when your instructor opens the workbook.

Submit for Exercise 2

- Worksheet 1: the same three measures on a relationship and on an inner join
- Worksheet 2: students by program from the unioned, cleaned, deduplicated raw files, left-joined to the lookup with “Unknown Program” visible
- Worksheet 3: a quality audit with at least five indicators
- A visible caption on each
- Packaged workbook (.twbx). No memo.